

The memory tier costs almost nothing to run.

What it has to buy is fewer attempts.

A workload model and unit economics for **MEMORY-SPEC.md**’s CockroachDB branch — closing the two numbers that file left open: what the tier costs at idle (\$10 decision 2), and whether it breaks **infra/README.md**’s “under \$1/month” claim (\$8).

Rates read from the AWS Price List Bulk API and vendor pricing pages on 2026-07-26 · volumes measured out of this repository, not assumed · generated by **infra/workload.py**

The answer, before the arithmetic

\$0.44 IDLE COST / MONTH, CORPUS AT REST The number MEMORY-SPEC §10.2 leaves open. CockroachDB adds \$0.00 of it. “Under \$1/month” survives the branch.	\$0.72 INFRASTRUCTURE / MONTH, T1 ACTIVE Whole stack — AWS, B2 and CockroachDB — while actually shipping 20 videos a month.	0.32% MEMORY TIER, AS SHARE OF A VIDEO \$0.0066 of memory against \$2.07 of generation and judging.	0.0075 ATTEMPTS IT MUST REMOVE TO PAY FOR ITSELF Per still, out of 2.2. Anything the memory tier actually learns clears this by three orders of magnitude.
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The finding that matters. This branch was written expecting a cost argument — MEMORY-SPEC §8 concedes it “reintroduces the tier that decision removed” and declines to restate the idle number. There is no cost argument to have. CockroachDB Basic scales to zero, publishes a 50M RU + 10 GiB monthly free allowance, and carries the distributed vector index on that plan. The workload fits inside it at every tier modelled here. **The real constraint is not price — it is that the vector-index RU cost is unpublished, so the free allowance has to be measured rather than assumed.**

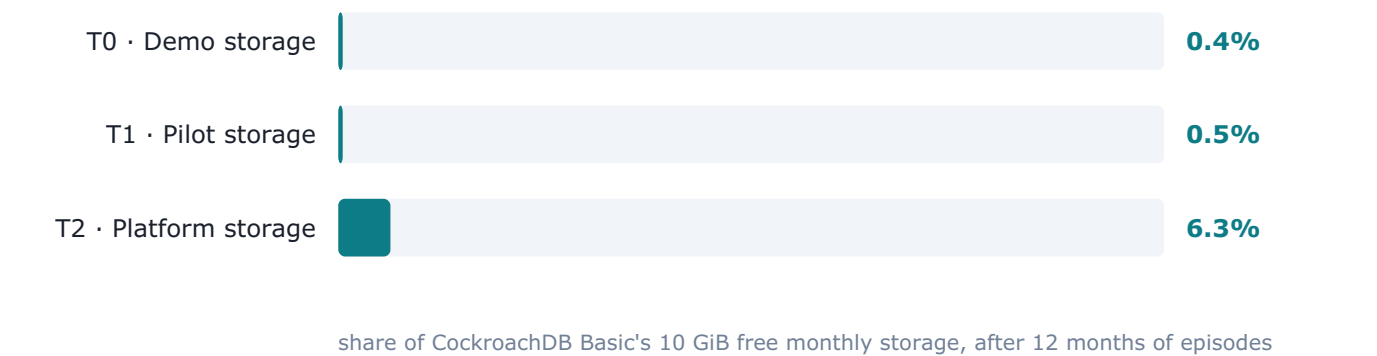
Where the work actually comes from

Volumes measured from the repo on 2026-07-26: 28 library clip descriptors, 22 payoff stills per edit across 5 existing edits, 5 reference framings per artist, and a mean of 47 tokens of embeddable prose per descriptor (p95 76).

DRIVER	T0 · DEMO	T1 · PILOT	T2 · PLATFORM
Tenants	1	1	10
Artists (identities / molds)	2	5	50
Songs	1	50	500
Clip corpus — vector-indexed rows	5,028	2,000	50,000
Edits per month	5	20	200
Stills per month	110	440	4,400
Attempts per approved still	3.0	2.2	1.8
Generation attempts per month	330	968	7,920
Likeness judgements per month	660	1,936	15,840
Memory-tier operations per month	1,750	5,240	43,600

T0 · Demo What exists on 2026-08-18, plus the synthetic corpus MEMORY-SPEC §9 commits to for exercising the index. Every count here is measured out of the repo.	T1 · Pilot Respect the Funk as tenant #1 at the scale PRODUCT.md names as the real bottleneck: fifty songs, where manual onboarding — not compute — throttles.	T2 · Platform Ten labels on one engine. The multi-tenant case BUILD-SPEC §2b rule 6 partitions for, and the first tier where the free allowance is a real question.
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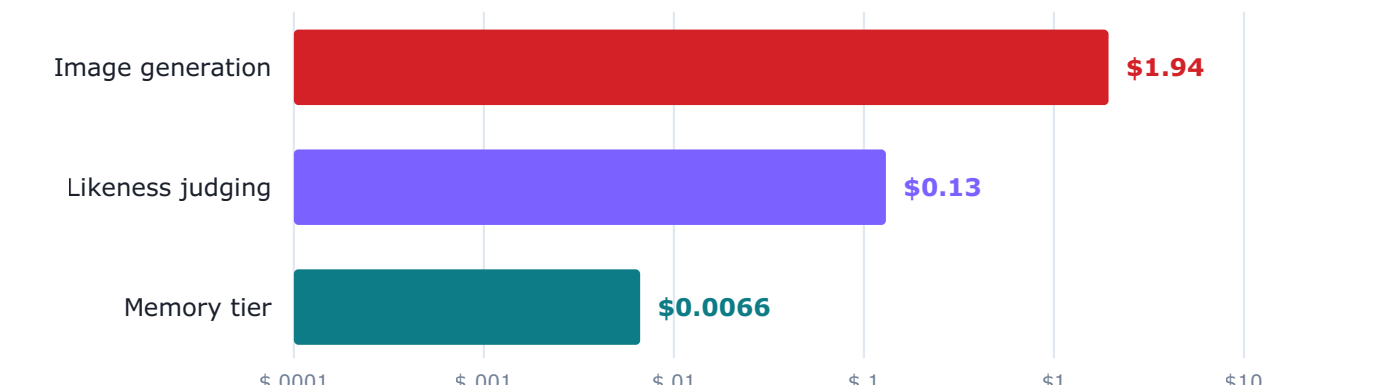
Free-allowance headroom — the only thing that can make this cost money



Storage is a non-event. A clip row is ~6.5 KB (a 2310-byte descriptor plus a 4 KB VECTOR(1024)); an episode row is ~3.6 KB. Even T2 — fifty thousand clips plus a full year of episodes — is 0.63 GiB, or 6.3% of the free 10 GiB. Bytes stay in B2; the tier holds vectors and rows, exactly as [memory-branch.pdf](#) draws it.

Request Units are the open number, and they are stated as an inversion rather than guessed. CockroachDB publishes no RU cost for a vector-index scan. So instead of inventing one: an operation would have to cost more than **9,542 RU** at T1 (**1,147 RU** at T2) before the 50M free allowance breaks. For scale, CockroachDB documents a small point read at single-digit RUs. T1 has room to spare; **T2 is where this must actually be measured** — day 1–2 of MEMORY-SPEC §9 is the place to record it.

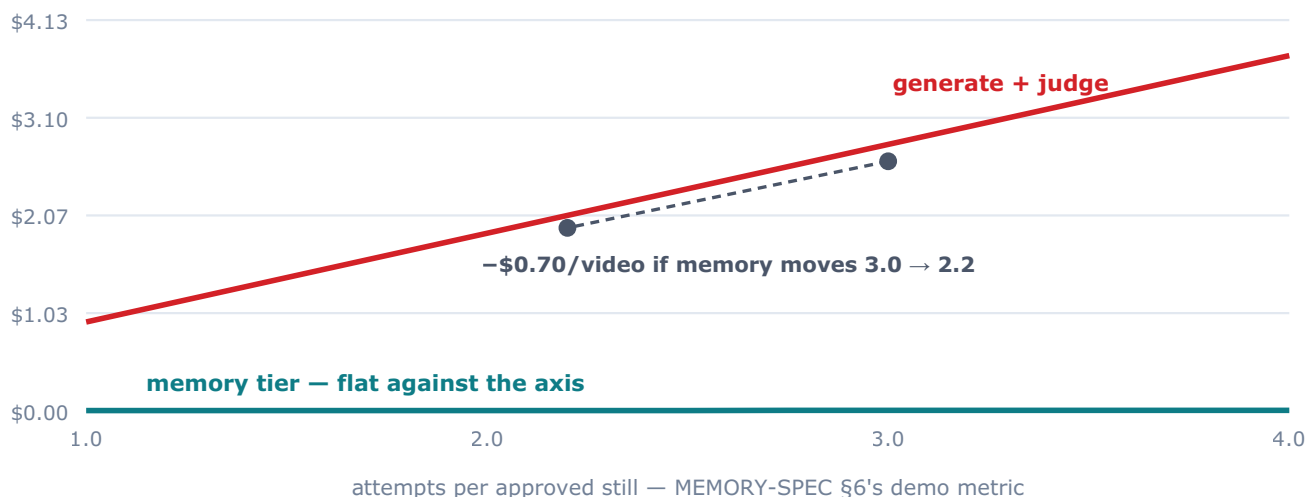
Unit economics — the memory tier is a rounding error on its own bill



Cost of one finished video at T1, on a log axis, because the ratio is the finding and a linear axis would render the teal bar invisible.

PER FINISHED VIDEO (22 STILLS)	T0 · DEMO	T1 · PILOT	T2 · PLATFORM
Image generation + likeness judging	\$2.82	\$2.07	\$1.69
Memory tier — embeddings, vectors, rows	\$0.0078	\$0.0066	\$0.0061
Memory tier as % of the video	0.28%	0.32%	0.36%
Cost per approved still	\$0.128	\$0.094	\$0.077
Attempts/still the tier must remove to break even	0.0089	0.0075	0.0069

The metric the whole case rests on



The trade, stated plainly. Memory costs \$0.0066 per video. One attempt-per-still of improvement is worth \$0.88. The tier therefore pays for itself at **0.0075** attempts removed per still — about 0.34% of the current figure. A move from 3.0 to 2.2 attempts saves \$0.70 per video, or **\$140.80/month at T2** — for a tier that bills nothing.

And the number this document cannot supply. Attempts-per-approved-still is *not measured anywhere in this repo*, because nothing writes back — that is precisely MEMORY-SPEC §3's third failure. The 3.0 / 2.2 / 1.8 figures above are **assumptions used to size the workload, not evidence**. The curve is drawn so the sensitivity is visible instead of hidden in a point estimate: read it as “here is what any improvement is worth”, never as a claim that the improvement has been observed. MEMORY-SPEC §6 already commits to labelling the demo curve with its N; the same rule governs this page.

What this does and does not change about the bill

- **research/07-cost-model's conclusion is unchanged and is reinforced.** Generation is the bill; everything else rounds to noise. The memory tier does not add a meaningful line — it is a lever on the line that already dominates.
- **The tier has no idle floor**, which is the property infra/README.md's no-database decision was actually protecting. Basic scales to zero; the free allowance is monthly, not a trial.
- **Embedding is free in practice.** The measured payload is ~47 tokens of prose per clip, so embedding a 50,000-clip catalogue costs \$0.03 — once.
- **What is genuinely new spend** is Bedrock, and it is \$0.13/month at T1. Batch embedding halves the ingest half of that.

The AWS workload estimate

Line items in AWS Pricing Calculator's own export schema, so this can be diffed against a real calculator estimate rather than retyped. Also emitted as **memory-workload.csv**. **API** read from the AWS Price List Bulk API · **MEAS** measured from this repo · **UNVERIFIED** could not be confirmed against a primary source.

T0 · Demo — 5 edits/month, 5,028 indexed clips, 1 tenant(s)

SERVICE	LINE	SPECS	\$/MO	\$/YR
Amazon Bedrock	Titan Text Embeddings V2 API	\$0.02/MTok · clip + query embeddings	\$0.0012	\$0.01
Amazon Bedrock	Titan Image Embeddings API	\$0.00006/image · 330 Q3 vectors/mo	\$0.0198	\$0.24
Amazon Bedrock	Nova Lite — agent runtime API	\$0.06/\$0.24 per MTok · 50 Q4 queries/mo	\$0.0180	\$0.22
AWS Batch / Fargate Spot	generator container SPOT	1.2 hr/mo · 2 vCPU / 4 GB · minvCpus 0	\$0.0370	\$0.44
AWS Lambda	web — Function URL API	1,000 req/mo — inside 1M free	\$0.0000	\$0.00
Amazon SQS	job queue + DLQ API	50 req/mo — inside 1M free	\$0.0000	\$0.00
Amazon ECR	image storage API	2 GB @ \$0.10/GB-mo	\$0.2000	\$2.40
Amazon CloudWatch	logs, 14-day UNVERIFIED	placeholder — infra/README.md carries this as “~cents”	\$0.1500	\$1.80
AWS Systems Manager	Parameter Store	Standard tier — free	\$0.0000	\$0.00
AWS subtotal			\$0.4260	\$5.11
CockroachDB Cloud	Basic — vectors and rows, never bytes	0.04 GiB of 10 GiB free · 1,750 ops/mo vs 50M free RU	\$0.0000	\$0.00
Backblaze B2	masters, stills, manifests — unchanged	2.8 GB @ \$0.00695/GB-mo	\$0.0191	\$0.23
Total infrastructure			\$0.4451	\$5.34
memo	image generation — gemini-2.5-flash-image MEAS	\$0.04/image · 330 attempts/mo	\$13.20	\$158.40

T1 · Pilot — 20 edits/month, 2,000 indexed clips, 1 tenant(s)

SERVICE	LINE	SPECS	\$/MO	\$/YR
Amazon Bedrock	Titan Text Embeddings V2 API	\$0.02/MTok · clip + query embeddings	\$0.0027	\$0.03
Amazon Bedrock	Titan Image Embeddings API	\$0.00006/image · 968 Q3 vectors/mo	\$0.0581	\$0.70
Amazon Bedrock	Nova Lite — agent runtime API	\$0.06/\$0.24 per MTok · 200 Q4 queries/mo	\$0.0720	\$0.86
AWS Batch / Fargate Spot	generator container SPOT	5.0 hr/mo · 2 vCPU / 4 GB · minvCpus 0	\$0.1481	\$1.78
AWS Lambda	web — Function URL API	4,000 req/mo — inside 1M free	\$0.0000	\$0.00
Amazon SQS	job queue + DLQ API	200 req/mo — inside 1M free	\$0.0000	\$0.00
Amazon ECR	image storage API	2 GB @ \$0.10/GB-mo	\$0.2000	\$2.40
Amazon CloudWatch	logs, 14-day UNVERIFIED	placeholder — infra/README.md carries this as “~cents”	\$0.1500	\$1.80
AWS Systems Manager	Parameter Store	Standard tier — free	\$0.0000	\$0.00
AWS subtotal			\$0.6309	\$7.57

CockroachDB Cloud	Basic — vectors and rows, never bytes	0.05 GiB of 10 GiB free · 5,240 ops/mo vs 50M free RU	\$0.0000	\$0.00
Backblaze B2	masters, stills, manifests — unchanged	13.2 GB @ \$0.00695/GB-mo	\$0.0918	\$1.10
Total infrastructure			\$0.7227	\$8.67
memo	image generation — gemini-2.5-flash-image MEAS	\$0.04/image · 968 attempts/mo	\$38.72	\$464.64

T2 · Platform — 200 edits/month, 50,000 indexed clips, 10 tenant(s)

SERVICE	LINE	SPECS	\$/MO	\$/YR
Amazon Bedrock	Titan Text Embeddings V2 API	\$0.02/MTok · clip + query embeddings	\$0.0242	\$0.29
Amazon Bedrock	Titan Image Embeddings API	\$0.00006/image · 7,920 Q3 vectors/mo	\$0.4752	\$5.70
Amazon Bedrock	Nova Lite — agent runtime API	\$0.06/\$0.24 per MTok · 2,000 Q4 queries/mo	\$0.7200	\$8.64
AWS Batch / Fargate Spot	generator container SPOT	50.0 hr/mo · 2 vCPU / 4 GB · minvCpus 0	\$1.4811	\$17.77
AWS Lambda	web — Function URL API	400,000 req/mo — inside 1M free	\$0.0000	\$0.00
Amazon SQS	job queue + DLQ API	2,000 req/mo — inside 1M free	\$0.0000	\$0.00
Amazon ECR	image storage API	2 GB @ \$0.10/GB-mo	\$0.2000	\$2.40
Amazon CloudWatch	logs, 14-day UNVERIFIED	placeholder — infra/README.md carries this as “~cents”	\$0.1500	\$1.80
AWS Systems Manager	Parameter Store	Standard tier — free	\$0.0000	\$0.00
AWS subtotal			\$3.0505	\$36.61
CockroachDB Cloud	Basic — vectors and rows, never bytes	0.63 GiB of 10 GiB free · 43,600 ops/mo vs 50M free RU	\$0.0000	\$0.00
Backblaze B2	masters, stills, manifests — unchanged	132.1 GB @ \$0.00695/GB-mo	\$0.9181	\$11.02
Total infrastructure			\$3.9686	\$47.62
memo	image generation — gemini-2.5-flash-image MEAS	\$0.04/image · 7,920 attempts/mo	\$316.80	\$3,801.60

Read the two totals together. Infrastructure at T2 is \$3.97/month while generation is \$316.80/month — infrastructure is 1.2% of the bill. That ratio is the whole reason the memory tier is worth adding despite the cost conflict MEMORY-SPEC §8 raises: it is spending in the free column to buy reductions in the expensive one.

What is not verified, and what would change these numbers

- **UNVERIFIED CockroachDB overage rates.** Basic's free allowance and scale-to-zero are on the vendor pricing page; the per-RU and per-GiB rates past it are third-party. They only matter if the free tier breaks, which is why every headline here is stated as headroom rather than as spend.
- **UNVERIFIED RU cost of a filtered vector scan.** The single largest modelling risk. Not published anywhere; must be measured on days 1–2 and is the cheapest possible thing to measure.
- **UNVERIFIED Fargate Spot discount.** Modelled at 70% off on-demand. Spot rates are dynamic and absent from the Price List API. At 5.0 hr/month this cannot move the total.
- **The face embedding is still an open design decision** (MEMORY-SPEC §10.3), priced here as Titan Image Embeddings at \$0.00006/image — a metric vector for Q3 rather than the vision model's judgement. A real change to `check_likeness.py`, not a refactor.
- **Attempts-per-approved-still is an assumption at every tier.** Restated here because it is the one number that would change the conclusion, and the one the repo cannot currently produce.

Sources — AWS Price List Bulk API (pricing.us-east-1.amazonaws.com, AmazonBedrock / AWSLambda / AWSQueueService / AmazonECS / AmazonECR, us-east-1, read 2026-07-26) · cockroachlabs.com/pricing/new and [/docs/cockroachcloud/costs](https://docs.cockroachcloud.com/costs) (read 2026-07-26) · [research/07-cost-model](#) (verified 2026-07-15) for Backblaze B2 and Gemini token rates · [content/bin/generate_stills.py](#) PRICES_USD (dated 2026-07 by the repo) for the image-generation rate.